

Computer Science & Information Technology

Discrete Mathematics

DPP: 3

Graph Theory

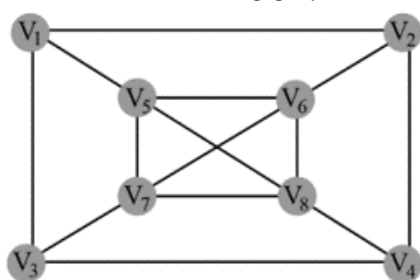
Q1 Which of the following options is/are correct?

- (A) The chromatic number of a graph with at least 1 edge is at least 2.
- (B) A graph is null graph if and only if its chromatic number is 1.
- (C) For any graph G , chromatic number of $G \leq 1 + \Delta(G)$.
- (D) For any graph G , chromatic number of $G \leq |V(G)|$

Q2 Which of the following options is/are correct?

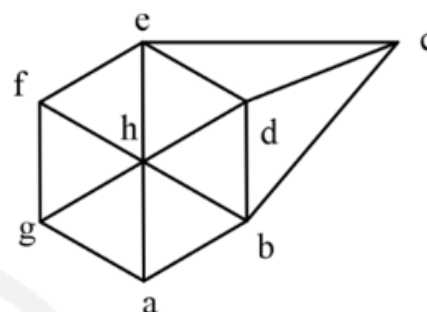
- (A) If chromatic number of graph G is 'm', then graph G is a m -partite graph, but it can never be (less than m)-partite.
- (B) If graph G is not a null graph and all the cycles of graph G are of even length, then chromatic number of G is 2.
- (C) If graph G is not a null graph and all the cycles of graph G are of even length, then G is a bipartite graph.
- (D) A tree with at least one edge is always a bipartite graph.

Q3 Consider the following graph,



Minimum number of edges that must be deleted from the graph such that the graph becomes a bipartite graph is _____

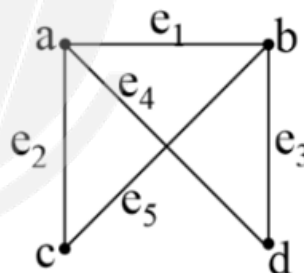
Q4 For the graph G shown below



Chromatic number of G + matching number of G = _____.

- (A) 6
- (B) 7
- (C) 5
- (D) 8

Q5 Number of maximal matching in the graph shown below is _____?



Q6 Which of the following is not True?

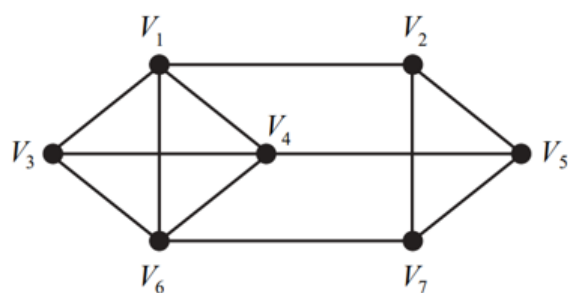
- (A) Number of perfect matching in $K_{2n} = \frac{(2n)!}{2^n n!}$
- (B) Number of perfect matching in $K_{n,n} = n!$
- (C) Number of perfect matching in C_n (n is even) = 2
- (D) Number of perfect matching in $w_{2n} = 2n$

Q7 Number of perfect matching in a tree with n -vertices ($n \geq 2$)

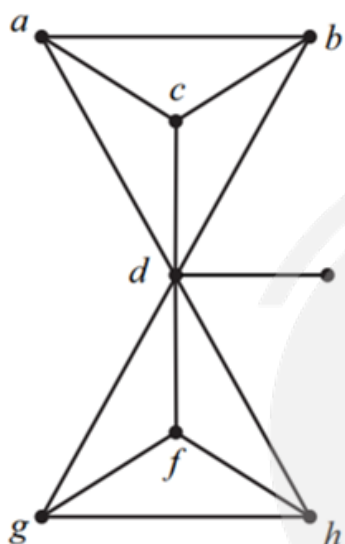
- (A) > 1
- (B) ≥ 1
- (C) ≤ 1
- (D) $= 1$

Q8 For the graph shown below, the chromatic number is _____.


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Q9 Find the matching number of the following graph



Q10 Let G be a simple graph with 20 Vertices and 100 edges. The size of the minimum vertex cover of G is 8. Then, the size of the maximum independent set of G is _____

Q11 What is the size of the smallest MIS (Maximal Independent Set) of a chain of nine nodes?

- (A) 5 (B) 4
(C) 3 (D) 2



Answer Key

Q1 (A, B, C, D)

Q2 (A, B, C, D)

Q3 2~2

Q4 (B)

Q5 3~3

Q6 (D)

Q7 (C)

Q8 4~4

Q9 3

Q10 12~12

Q11 (C)



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Hints & Solutions

Q1 Text Solution:

- (A) An edge connects two vertices and those two vertices have to be colored using two different colors (because they are adjacent).
 (B) A null graph has no edge, so no vertices are adjacent. Hence, each vertex can be colored with a single color.
 (C) It is a property of any graph.
 (D) In worst case, we might have to color each vertex of a graph with a different color which makes the maximum chromatic number of a graph $= |V(G)|$. Hence in general, chromatic number of a graph $\leq |V(G)|$.
 Hence, all options are correct.

Q2 Text Solution:

- (A) If chromatic number of a graph G is ' m ', then it signifies that there is a vertex in G that is adjacent to $m-1$ vertices. So, these m vertices will make m partitions and the graph will be a m -partite graph.
 (B) A cycle of even length can be colored alternately using two colors.
 (C) This statement is derived using option A and option B.
 (D) The chromatic number of a tree with at least one edge is always 2.

Hence, all options are correct.

Q3 Text Solution:

$V_5 V_8$ and $V_6 V_7$ are the two edges to be deleted.

Q4 Text Solution:

The three partitions can be $\{a, d, f\}$, $\{b, e, g\}$, and $\{c, h\}$. Hence, chromatic number is 3. One possible matching set can be $\{ec, db, ga, fh\}$. Hence, matching number is 4.

Q5 Text Solution:

There are three possible maximal matching $\{e_1\}$, $\{e_2, e_3\}$, and $\{e_4, e_5\}$.

Q6 Text Solution:

The number of perfect matchings in W_{2n} is $2n-1$.

Q7 Text Solution:

A tree with an odd number of vertices cannot have a perfect matching because there will always be one vertex left unmatched.
 If n is even, the number of perfect matchings can be 1 or more depending on the tree's structure.

Q8 Text Solution:

The four partitions can be $\{V_1, V_5\}$, $\{V_2, V_3\}$, $\{V_6\}$ and $\{V_4, V_7\}$.
 Hence, chromatic number is 4.

Q9 Text Solution:

One possible matching set can be $\{ad, bc, gf\}$.
 Hence, matching number is 3.

Q10 Text Solution:

Total no. of vertices $= VC(G) + IS(G)$
 $\Rightarrow 20 = 8 + IS(G)$
 $\Rightarrow IS(G) = 20 - 8 = 12$

Q11 Text Solution:

$a_1 - \textcircled{a_2} - a_3 - a_4 - \textcircled{a_5} - a_6 - a_7 - \textcircled{a_8} - a_9$



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